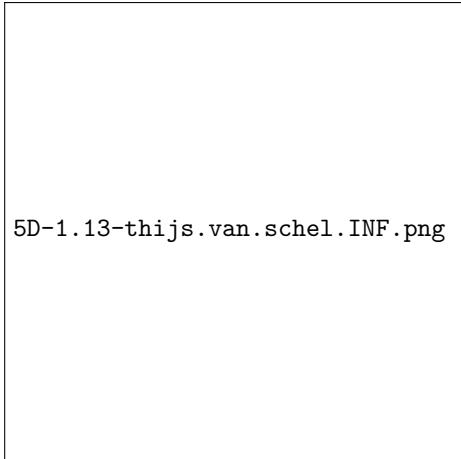


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Oefeningenreeks 5D nr. 1.13

$$\begin{aligned}f(x) &= \frac{x}{\sqrt{1+x^2}} \\&= \int_{-1}^{\sqrt{7}} \left| \frac{x}{\sqrt{1+x^2}} \right| dx \\&= \int_{-1}^0 \frac{-x}{\sqrt{1+x^2}} dx + \int_0^{\sqrt{7}} \frac{x}{\sqrt{1+x^2}} dx \\u &= 1+x^2, du = -2x dx \\&\Rightarrow \int \frac{1}{2\sqrt{u}} du = \sqrt{u} + C = \sqrt{1+x^2} + C \\&\Rightarrow \text{OPP} = [-\sqrt{1+x^2}]_{-1}^0 + [\sqrt{1+x^2}]_{\sqrt{7}}^0 \\&= -1 + \sqrt{2} + 2\sqrt{2} - 1 = 3\sqrt{2} - 2\end{aligned}$$

Tekening:



5D-1.13-thijs.van.schel.INF.png

References